

Data Visualisation

Day 1 — From Data to Insight

Course Introduction · Data Fundamentals · Tools · Data Types

9-week programme · Excel · Google Sheets · Python

Agenda – Day 1

Morning

1. Data → Information → Insight
2. Why visualisation matters
3. The visualisation workflow
4. Our three tools

Afternoon

5. Data types – the full picture
6. Which data type → which chart?
7. Hands-on exercise
8. Q & A + homework

By end of day you will understand **what data is**, **what it becomes**, and **how we work with it** for the rest of this course.

01

What Is Data?

And Why Does It Matter?

Data · Information · Insight

A single transaction:

Invoice

Craigs' Landscaping
123 Sierra Dr
Palo Alto, CA 94028

123-456-7890
craigslandscaping@email.com
craigslandscaping.com

Bill To

Smith Enterprises
456 Sierra Dr
Palo Alto, CA 94028

Invoice Details

Invoice # 12345
Terms Net 30
Invoice date 08/01/2024
Due date 08/31/2024

Product/Services	Qty/Hrs	Rate	Amount
Stone work	3.4	\$125	\$425
Mulching, weeding, planting	3	\$75	\$225
Subtotal			\$650.00
Discount			-\$65.00
Tax 8.5%			\$49.73
Total			\$634.73

Table record each transaction:

Date	Invoice	Customer	Product	Qty	Unit Price	Total
2024-01-03	INV-001	Alice Johnson	Laptop X1	1	\$899	\$899
2024-02-14	INV-002	Bob Martinez	Mouse Pro	3	\$120	\$360
2024-03-07	INV-003	Carol White	Laptop X1	1	\$899	\$899
2024-01-22	INV-004	David Lee	Keyboard K2	2	\$150	\$300
2024-04-11	INV-005	Emma Brown	Monitor 27"	1	\$450	\$450
2024-05-03	INV-006	Frank Wilson	Mouse Pro	2	\$120	\$240
2024-02-28	INV-007	Grace Taylor	Laptop X1	1	\$899	\$899
2024-06-15	INV-008	Henry Clark	Monitor 27"	2	\$450	\$900
2024-03-19	INV-009	Irene Scott	Keyboard K2	1	\$150	\$150
2024-05-27	INV-010	James Rivera	Mouse Pro	5	\$120	\$600
...						

Data · Information · Insight

Table record each transaction:

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2024-04-11	INV-005	Emma Brown	Monitor 27"	1	\$450	\$450
2024-05-03	INV-006	Frank Wilson	Mouse Pro	2	\$120	\$240
2024-02-28	INV-007	Grace Taylor	Laptop X1	1	\$899	\$899
2024-06-15	INV-008	Henry Clark	Monitor 27"	2	\$450	\$900
2024-03-19	INV-009	Irene Scott	Keyboard K2	1	\$150	\$150
2024-05-27	INV-010	James Rivera	Mouse Pro	5	\$120	\$600
...						

Sum of transactions by month:

Month	Revenue
Jan	42
Feb	55
Mar	48
Apr	62
May	78
Jun	85

Data · Information · Insight

Data

Raw, unprocessed facts — no context, no meaning.

42, 87, 13, 56

2024-01-03, North, 5000

Information

Data **with context** — structured, labelled, comparable.

"Monthly revenue Jan–Apr: 42k, 87k, 13k, 56k USD"

Insight

Information that drives **action** — the "so what?" answer.

"Revenue collapsed in March — investigate supply chain."

Visualisation is the bridge that turns information into insight at a glance.

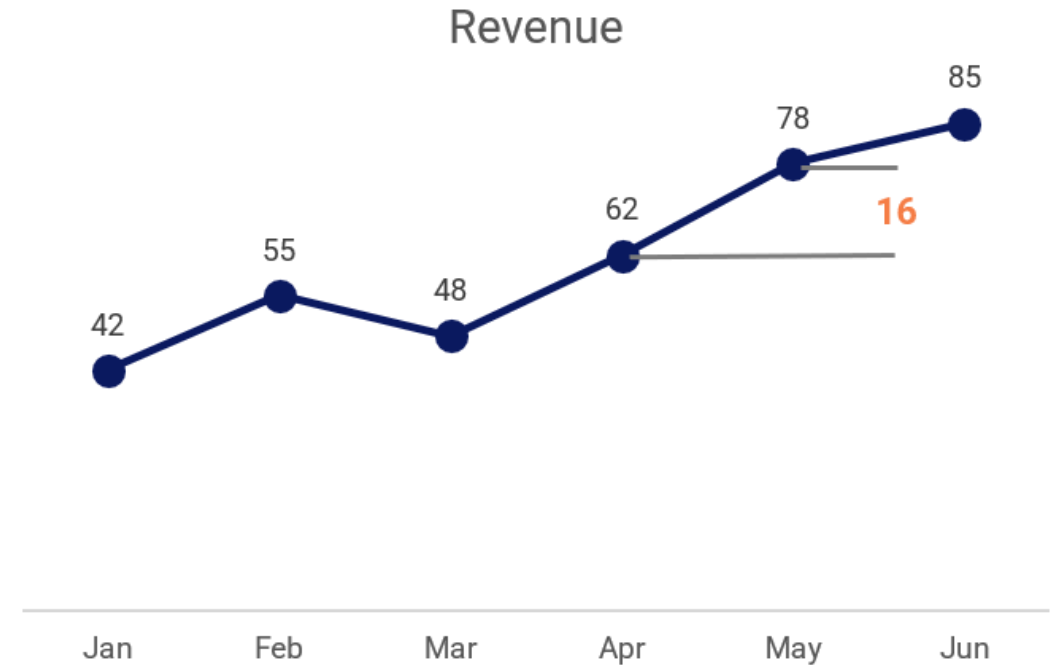
Why Visualisation Matters

Without a chart — find the biggest monthly jump:

Month	Revenue
Jan	42
Feb	55
Mar	48
Apr	62
May	78
Jun	85

How long did that take?

With a line chart — answer in under a second:



Apr → May (+16k) is the biggest jump.

The Visualisation Workflow

Every session in this course follows the same pipeline:



```
graph LR; A[Raw Data] --> B[Clean & Structure]; B --> C[Presentation Table]; C --> D[Chart / Dashboard]; D --> E[Story + Decision]
```

Raw Data ▶ Clean & Structure ▶ Presentation Table ▶ Chart / Dashboard ▶ Story + Decision

We cover **every step** — not just the chart at the end.

The table step is where most analysts lose quality.

Ex01

Raw Data ▶ Clean & Structure ▶ Presentation Table

Ronnykym. (2020). Online Store Sales Data [Data set]. Kaggle.

Use Excel formulas or pivot table to report on sale by month of the year 2019 and 2020.

Ex01

Sales by month (EUR)			
Month	2019	2020	pctChange
Jan	4,516,615	5,215,700	15%
Feb	2,895,564	1,959,413	-32%
Mar	2,521,828	3,579,944	42%
Apr	4,172,237	4,069,098	-2%
May	3,825,682	5,537,944	45%
Jun	7,692,235	7,158,138	-7%
Jul	6,638,609	5,242,556	-21%
Aug	3,437,450	4,714,525	37%
Sep	4,870,061	3,778,141	-22%
Oct	4,372,150	4,869,727	11%
Nov	2,961,319	4,734,730	60%
Dec	8,713,776	5,884,298	-32%
	56,617,524	56,744,214	0.22%

02

Our Tools

Excel · Google Sheets · Python

Three Tools, One Workflow

Microsoft Excel

- Business reports
- Quick prototypes
- PivotTables
- One-off analysis

Insert → **Charts wizard**

Google Sheets

- Shared dashboards
- Live data feeds
- Team collaboration
- Cloud access

Insert → **Chart panel**

Python

- Large datasets
- Automation & scripts
- Publication quality
- Reproducible analysis

`matplotlib`

`seaborn`

`plotly`

*Knowing **which tool to reach for** is itself a professional skill.*

Excel — Core Skills We Will Cover

Data work

- PivotTable — group, aggregate, pivot
- VLOOKUP / XLOOKUP — merge tables
- Conditional formatting
- Data validation

Charts

Line · Bar · Pie · Waterfall

Funnel · Stock (Candlestick) · Combo

Ex01.02

Raw transaction log

Date	Region	Revenue
Jan 3	North	5,000
Jan 5	South	4,500
...		

▼ PivotTable

Region	Q1	Q2
North	42k	55k
South	38k	47k

Google Sheets — When to Choose It

Strengths over Excel

- Real-time multi-user editing
- Built-in `QUERY()` — SQL-like filtering
- `IMPORTRANGE()` — live data from another sheet
- Free, browser-based, easy sharing
- Native connection to Google Data Studio

Situation	Choose
Colleagues editing simultaneously	Sheets
Data stays on your machine	Excel
Pulling live web / API data	Sheets
Complex macros / VBA	Excel
Sharing with non-Office users	Sheets
Large files (> 100MB)	Excel

Python – Three Libraries, Three Purposes

matplotlib

The **foundation**. Full manual control over every element.

```
import matplotlib.pyplot as plt

plt.plot(months, revenue,
         color='#0D1A63')
plt.title('Monthly Revenue')
plt.show()
```

seaborn

Statistical charts with minimal code. Built on matplotlib.

```
import seaborn as sns

sns.boxplot(
    x='Group',
    y='Score',
    data=df
)
```

plotly

Interactive charts – hover, zoom, filter. Web & dashboards.

```
import plotly.express as px

fig = px.bar(df,
             x='Month',
             y='Revenue')
fig.show()
```


03

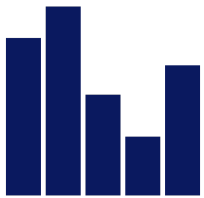
Chart Types by Purpose

Comparison · Composition · Relationship · Distribution

Chart Types — By Purpose

Comparison

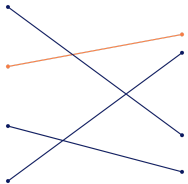
BAR



LINE

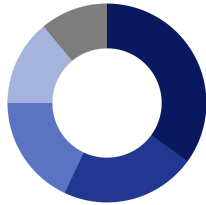


SLOPE

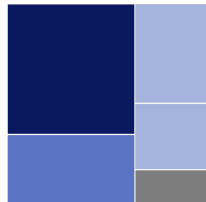


Composition

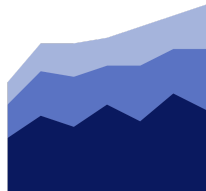
DONUT



TREEMAP

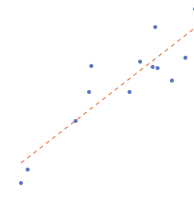


STACKED AREA

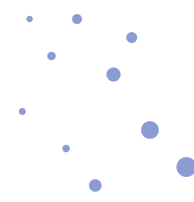


Relationship

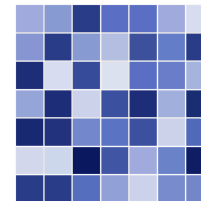
SCATTER



BUBBLE

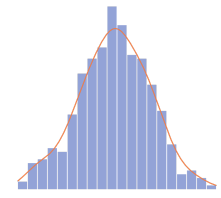


HEATMAP

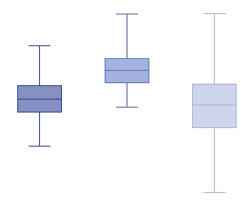


Distribution

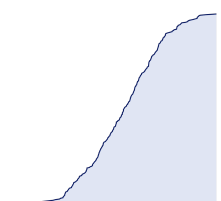
HISTOGRAM



BOX PLOT



CDF

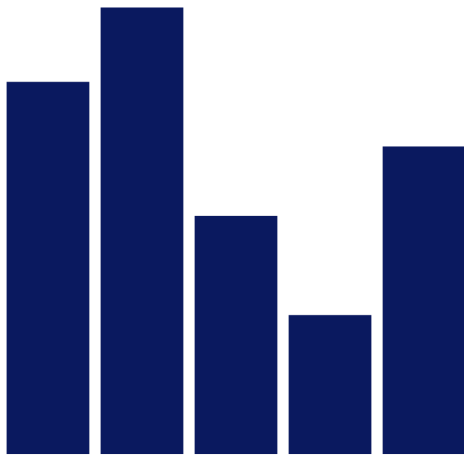


03

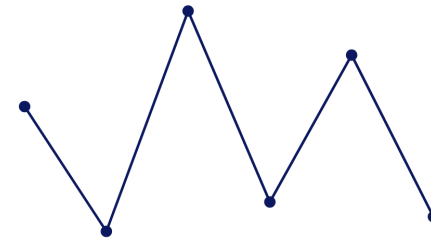
Chart Types by Function

Finance · Sales & Marketing · Science & Research

Finance Charts – Trends & Comparison

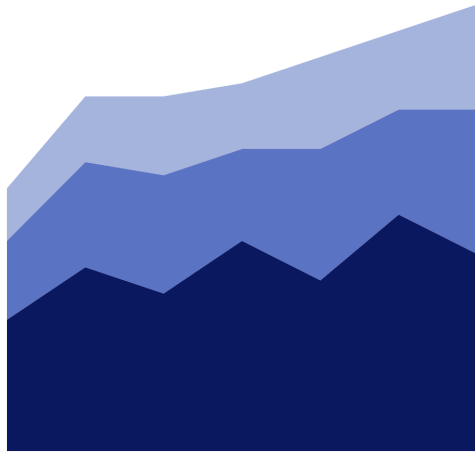


BAR – COMPARE REVENUE ACROSS PERIODS OR SEGMENTS

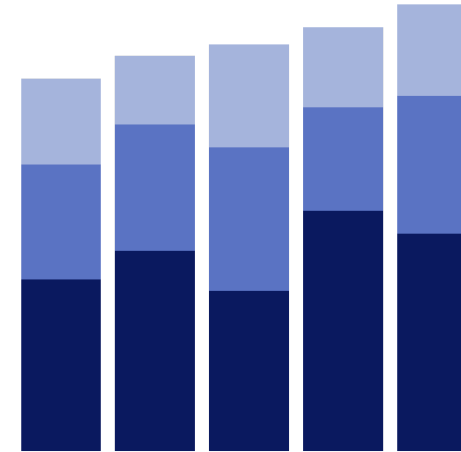


LINE – TRACK FINANCIAL METRICS OVER TIME

Finance Charts — Composition

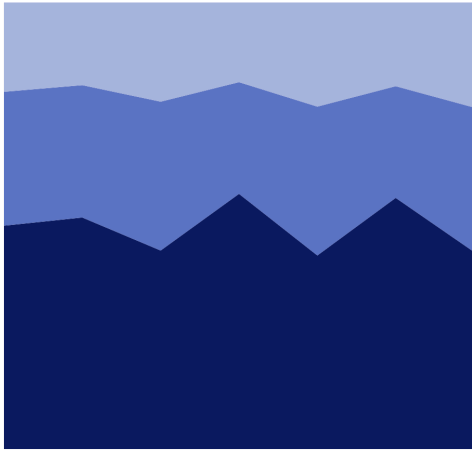


STACKED AREA — REVENUE MIX OVER TIME

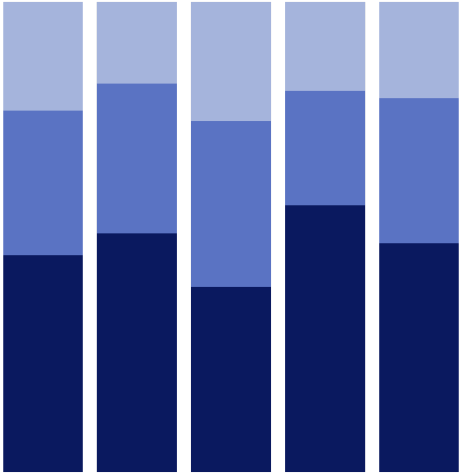


STACKED COLUMN — PERIOD BREAKDOWN BY CATEGORY

Finance Charts – Relative Composition



100% STACKED AREA – RELATIVE SHARE OVER TIME

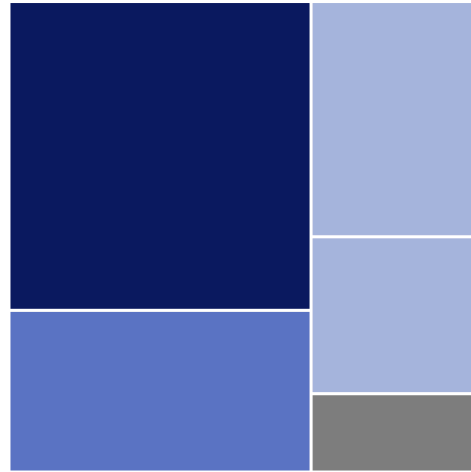


100% STACKED COLUMN – PROPORTIONAL BREAKDOWN PER PERIOD

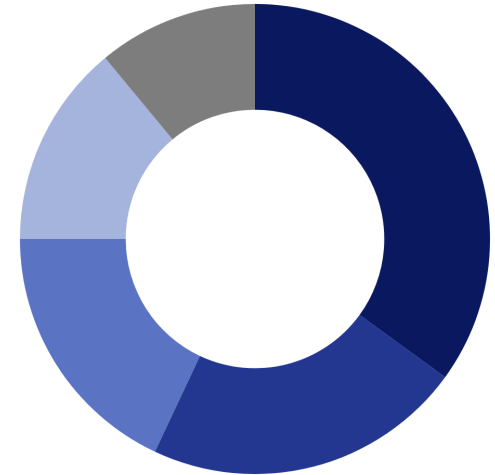
Finance Charts – KPI & Allocation



GAUGE – SINGLE KPI DIAL

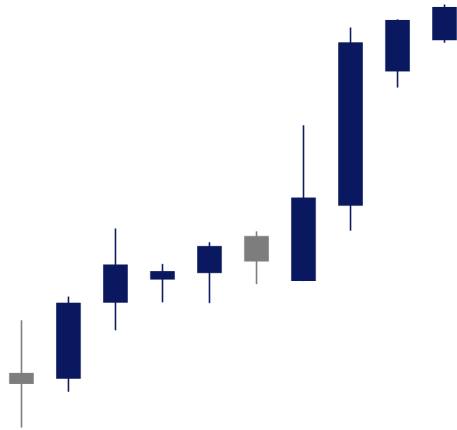


TREEMAP – PORTFOLIO / BUDGET
BREAKDOWN

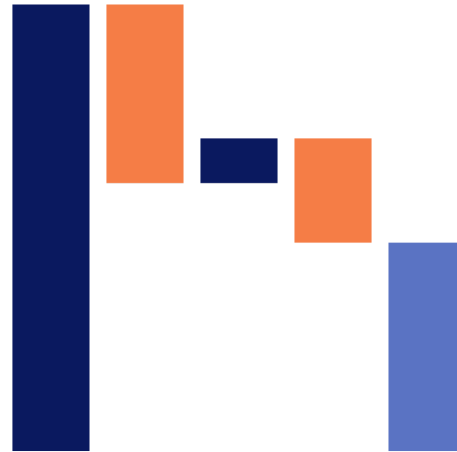


DONUT – ASSET / REVENUE MIX

Finance Charts — Prices & Performance



CANDLESTICK — OHLC STOCK PRICES

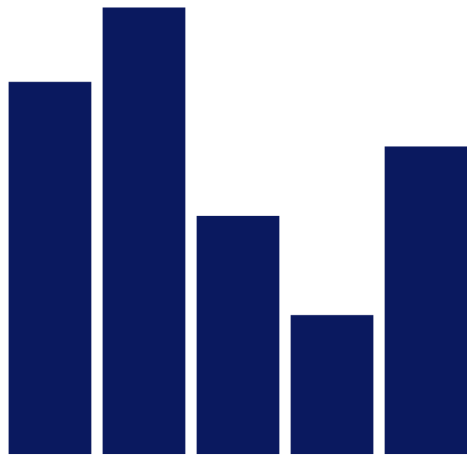


WATERFALL — P&L BRIDGE

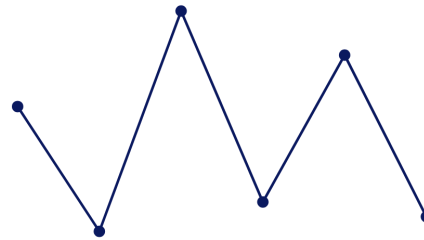


BULLET — KPI VS TARGET

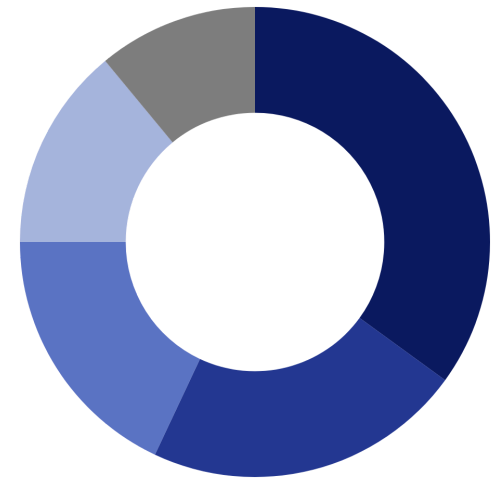
Sales & Marketing Charts – Trends & Mix



BAR – COMPARE ACROSS CATEGORIES

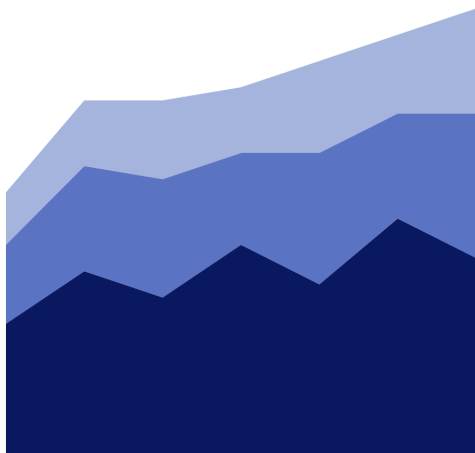


LINE – REVENUE TREND OVER TIME

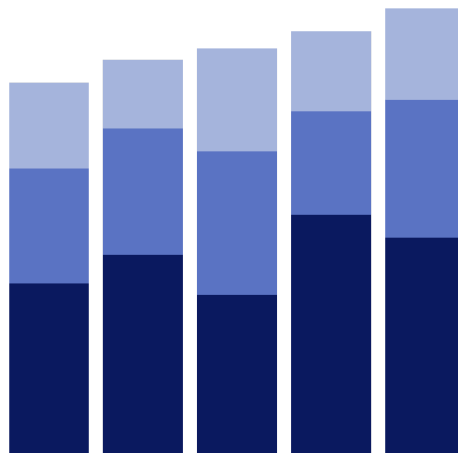


DONUT – MARKET SHARE / MIX

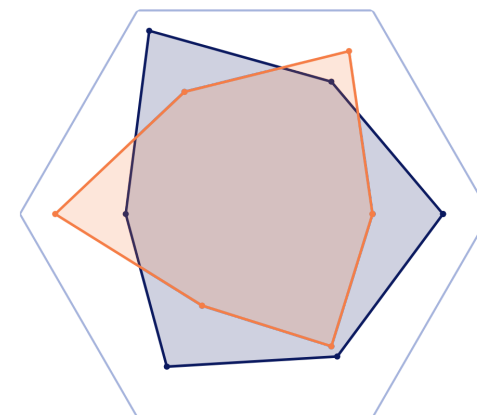
Sales & Marketing Charts – Composition



STACKED AREA – CHANNEL MIX TREND
OVER TIME



STACKED COLUMN – SALES BREAKDOWN BY
SEGMENT

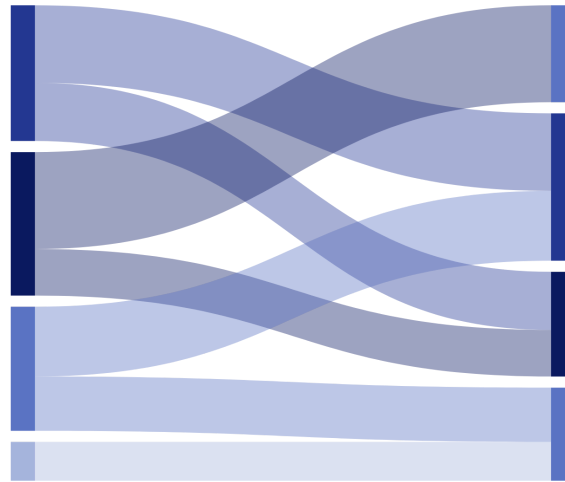


RADAR – MULTI-KPI PRODUCT / CAMPAIGN
COMPARISON

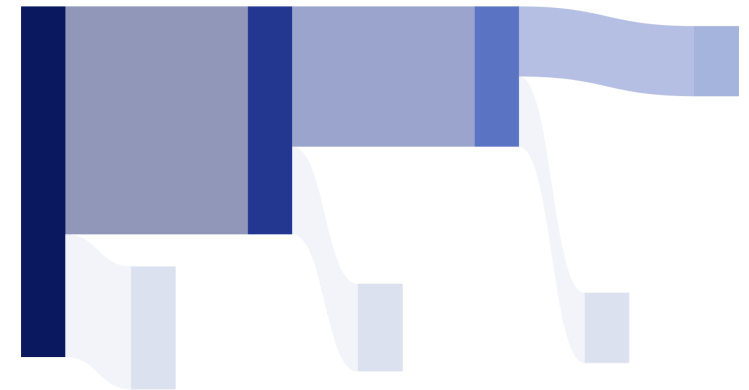
Sales & Marketing Charts – Flow & Pipeline



FUNNEL – CONVERSION PIPELINE

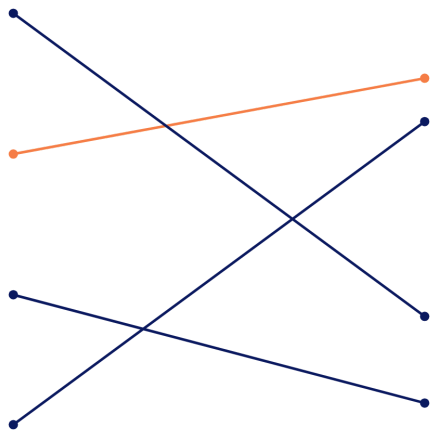


ALLUVIAL – CUSTOMER JOURNEY FLOWS

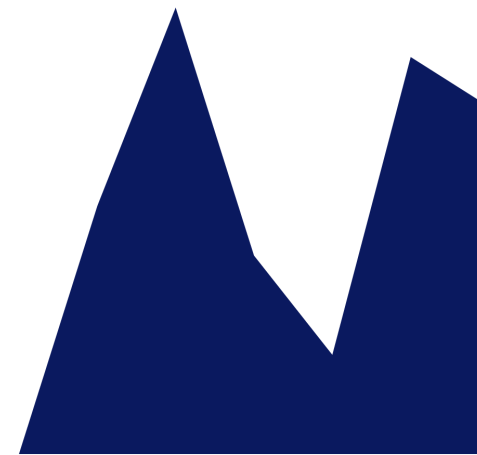


SANKEY – MULTI-LAYER DROP-OFF

Sales & Marketing Charts – Comparison

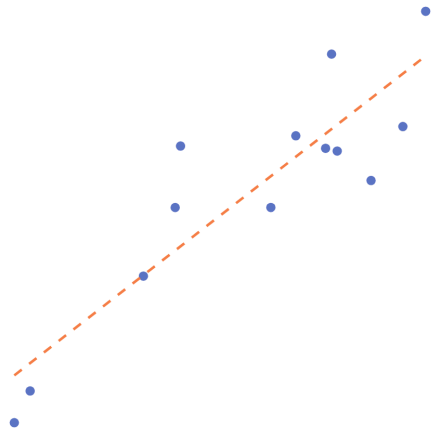


SLOPE – BEFORE VS AFTER

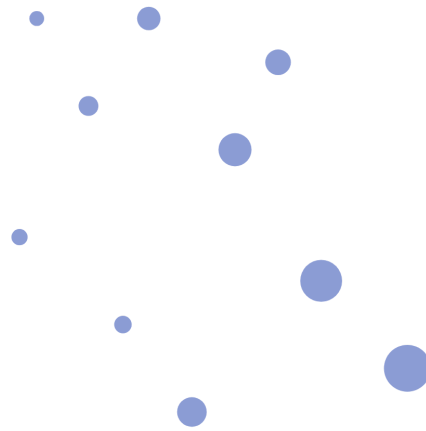


AREA – CUMULATIVE REVENUE OVER TIME

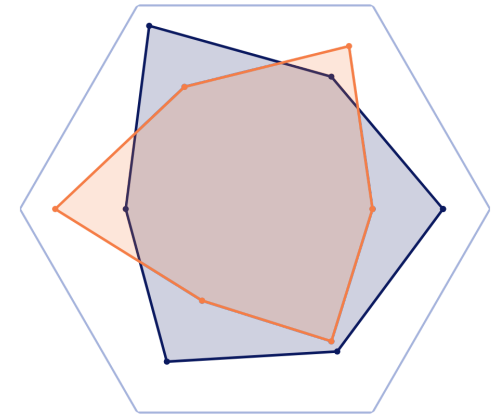
Science & Research Charts – Relationships



SCATTER – CORRELATION (2 VARIABLES)

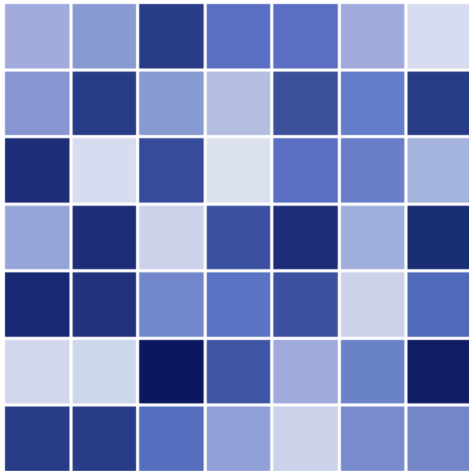


BUBBLE – 3-VARIABLE RELATIONSHIP

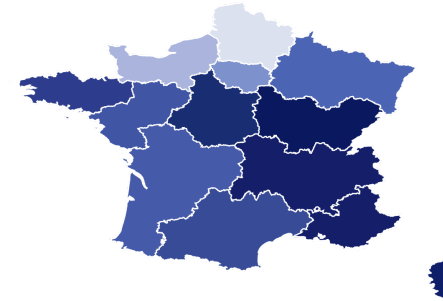


**RADAR – MULTI-ATTRIBUTE PROFILE
COMPARISON**

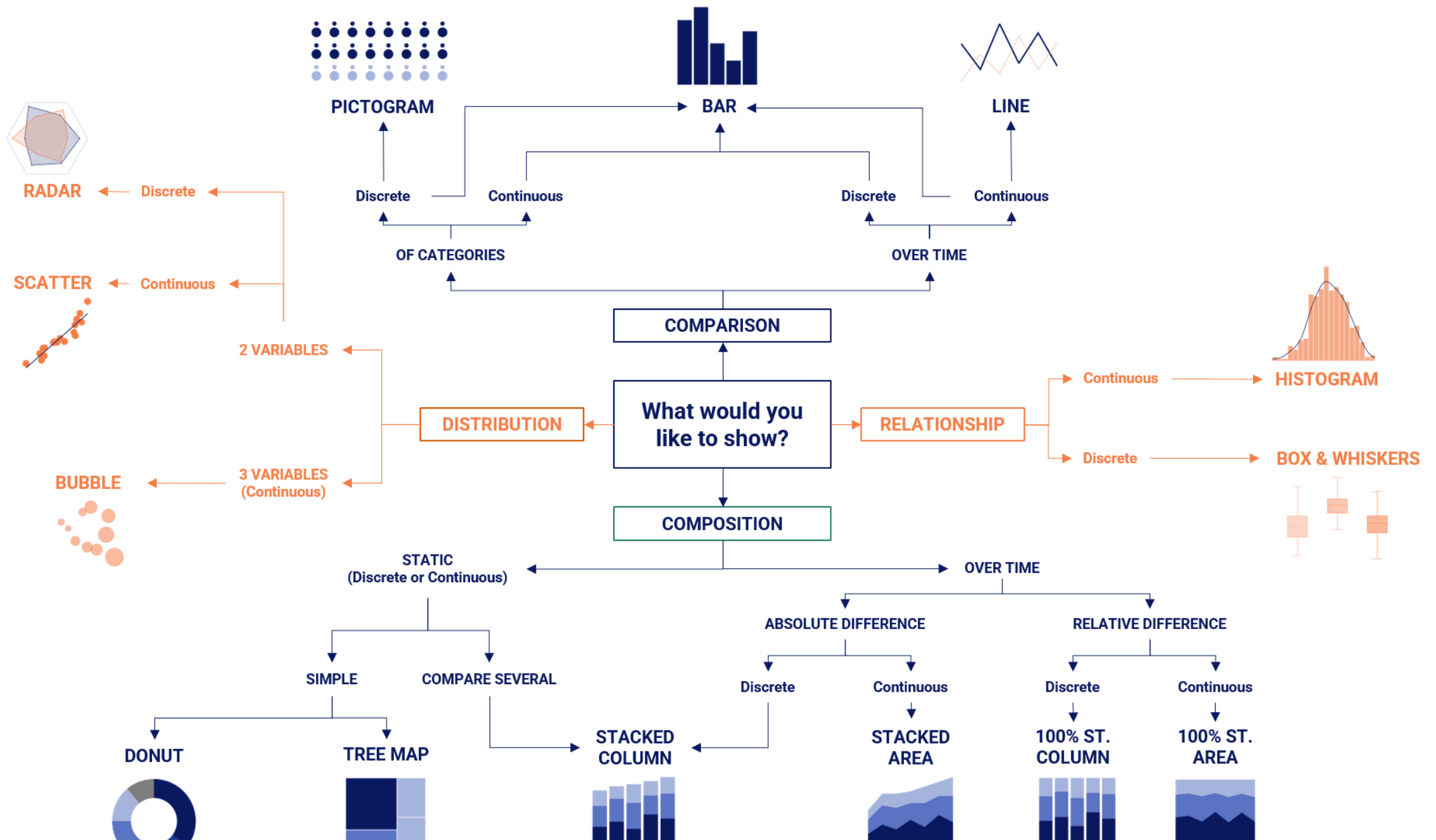
Science & Research Charts – Patterns & Geography



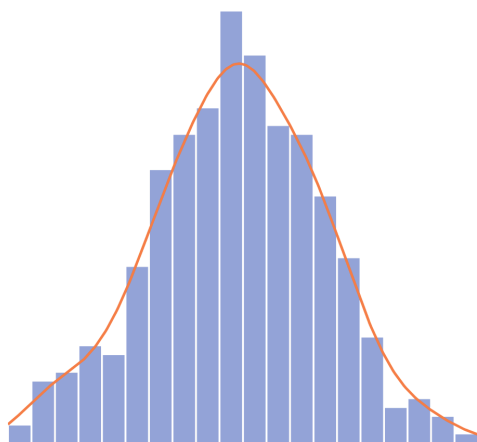
HEATMAP – PATTERNS ACROSS TWO DIMENSIONS



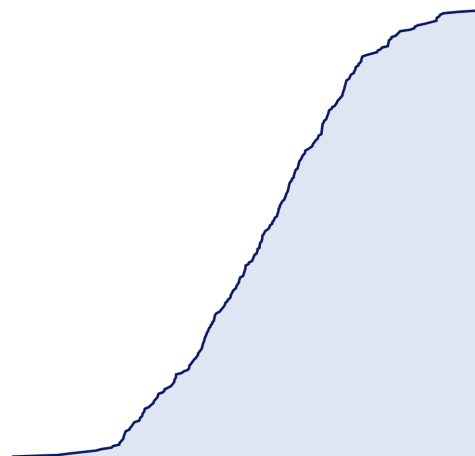
CHOROPLETH – METRIC BY REGION



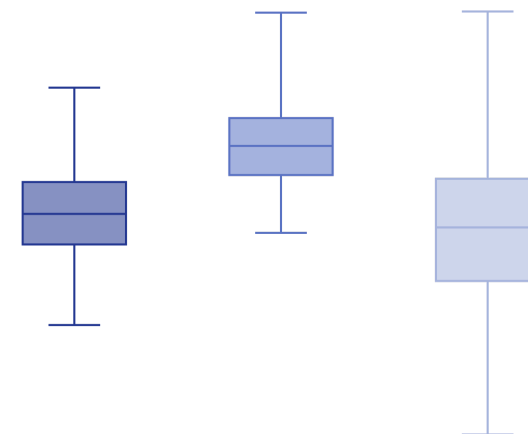
Science & Research Charts – Distributions



**HISTOGRAM + KDE – FREQUENCY
DISTRIBUTION WITH DENSITY CURVE**



**CDF – CUMULATIVE PROBABILITY OF A
DISTRIBUTION**



**BOX PLOT – DISTRIBUTION & OUTLIERS
ACROSS GROUPS**

Data Types

| A Important Concept in Visualisation

Why Data Types Matter

The data type determines the chart — not the other way around.

Trying to draw a line chart on nominal categories, or a pie chart on continuous data, produces charts that are meaningless — or actively misleading.

Data type	Examples	Suitable charts
Continuous	Revenue, temperature, height	Line, scatter, histogram
Discrete	Customers, units, page views	Bar, dot plot
Nominal	Country, product name, channel	Bar, pie, treemap
Ordinal	Rating: Poor / Good / Excellent	Bar, heatmap
Temporal	Dates, timestamps	Line, area, candlestick
Geospatial	City, coordinates, country	Map, choropleth

Quantitative – Continuous vs Discrete

Continuous

Can take **any value** in a range — infinitely many possible values exist between two points.

Examples

- Revenue: \$47,382.56
- Temperature: 36.8°C
- Height: 1.73 m
- Profit margin: 12.4%

Key charts

Line · Scatter · Histogram · Box Plot

Discrete

Countable whole numbers only.
You cannot have 2.5 customers.

Examples

- Customers: 847
- Units sold: 1,200
- Page views: 54,321
- Support tickets: 7

Key charts

Bar · Column · Dot Plot

Categorical – Nominal vs Ordinal

Nominal

Categories with **no natural order**.

You cannot say North > South.

Examples

- Region: North / South / East / West
- Product: Widget A / Widget B
- Channel: Email / Social / Search
- Country: Vietnam / USA / UK

Key charts

Bar · Pie · Donut · Treemap

Ordinal

Categories **with a meaningful order** — but the gap between levels may not be equal.

Examples

- Rating: Poor · Fair · Good · Excellent
- Education: High School · Bachelor's · Master's
- Priority: Low · Medium · High · Critical
- NPS: Detractor · Passive · Promoter

Key charts

Bar (sorted) · Heatmap · Stacked Bar

Temporal – Time-based Data

Data recorded at **points in time** or over **time intervals**.

Regular intervals

Daily stock prices, monthly sales, hourly temperature readings.

The gap between points is constant — patterns like seasonality and trends become visible.

Charts: Line · Area · Candlestick · Calendar Heatmap

Irregular intervals

Customer purchase timestamps, server log events, support tickets.

Points are unevenly spaced — aggregation (hourly → daily → monthly) is needed first.

Charts: Line after aggregation · Event timeline

Golden rule: Time always goes on the **X-axis**. The metric goes on the **Y-axis**.

Geospatial Data

Data with a **location** component.

Types of location data

Type	Example
Country name	"Vietnam", "USA"
Region / Province	"North", "HCMC"
Coordinates	lat 10.8°, lon 106.7°
Postal code	70000
Address	"123 Nguyen Hue St"

Key charts

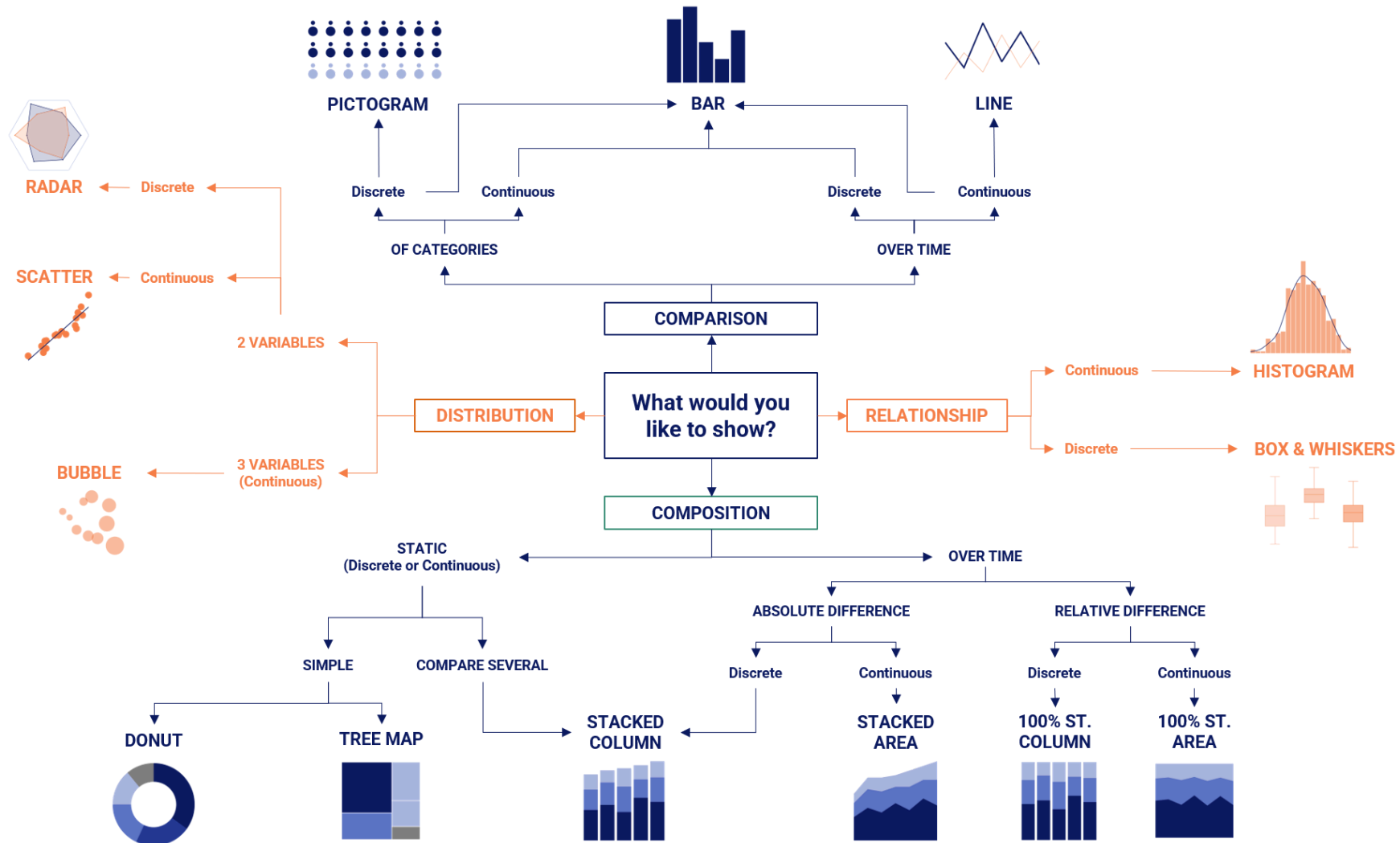
Choropleth map — colour regions by value (sales by country)

Dot / bubble map — sized dots at coordinates (store locations)

Geo heatmap — density of events (customer clusters)

We cover choropleth maps using `plotly` in Part 2.

The Data-Type Decision Tree



04

Hands-On

Identify Data Types in a Real Dataset

Exercise — Label Every Column

Look at this dataset. Working in pairs, label each column with its data type.

Date	Salesperson	Region	Product	Units Sold	Unit Price	Revenue
Jan 03	Alice	North	Widget A	50	100	5,000
Jan 05	Bob	South	Widget B	30	150	4,500
Jan 12	Carol	East	Widget A	40	100	4,000

Discuss for 3 minutes:

- Which columns are numerical? Continuous or discrete?
- Which are categorical? Nominal or ordinal?
- Which is temporal?
- What charts could we build from this data?

Exercise — Answers

Column	Data Type	Reason
Date	Temporal	Ordered time series — trend charts possible
Salesperson	Nominal	Names, no natural rank order
Region	Nominal	Locations, no natural rank order
Product	Nominal	Category labels, no order
Units Sold	Discrete	Whole numbers only, countable
Unit Price	Continuous	Can be any decimal value
Revenue	Continuous	Derived from Units × Price, any decimal

Charts we can build:

Revenue over time (line) · Revenue by region (bar) · Product mix (pie) · Units vs Price (scatter)

Day 1 — Key Takeaways

Core concepts

- **Data** = raw facts, no meaning alone
- **Information** = data with context
- **Insight** = information that drives action
- Visualisation accelerates the journey

Data types

Continuous · Discrete · Nominal · Ordinal · Temporal ·
Geospatial

Data type determines chart — not personal preference.

Tools

- **Excel** — fast, offline, business standard
- **Sheets** — collaborative, cloud, live data
- **Python** — powerful, automated, reproducible
 - `matplotlib` foundations
 - `seaborn` statistics
 - `plotly` interactive

Workflow

Raw Data → Clean Table → Chart → Insight

Homework Before Day 2

1. Data types in the wild

Find any report, news article, or dashboard. Label every column or axis label with its data type and write two sentences explaining your choices.

2. Environment check

Activate the `venv` and run:

```
import matplotlib; print(matplotlib.__version__)
```

3. Optional reading

Storytelling with Data — Chapter 1: "The importance of context"

See You on Day 2

Finance Visualisation

Next session covers:

Trial Balance · P&L Statement · Balance Sheet · Waterfall Chart · Candlestick Chart

Questions before Day 2? Reach out anytime.

Sources consulted: [CleanChart](#) — Financial Data Visualization · [Julius AI](#) — Finance Visualization · [HubSpot](#) — Chart Types Guide · [SR Analytics](#) — Visualization Techniques · [Yale Library](#) — Data Visualization Types · [Scientific Discovery](#) — Visualization Guide